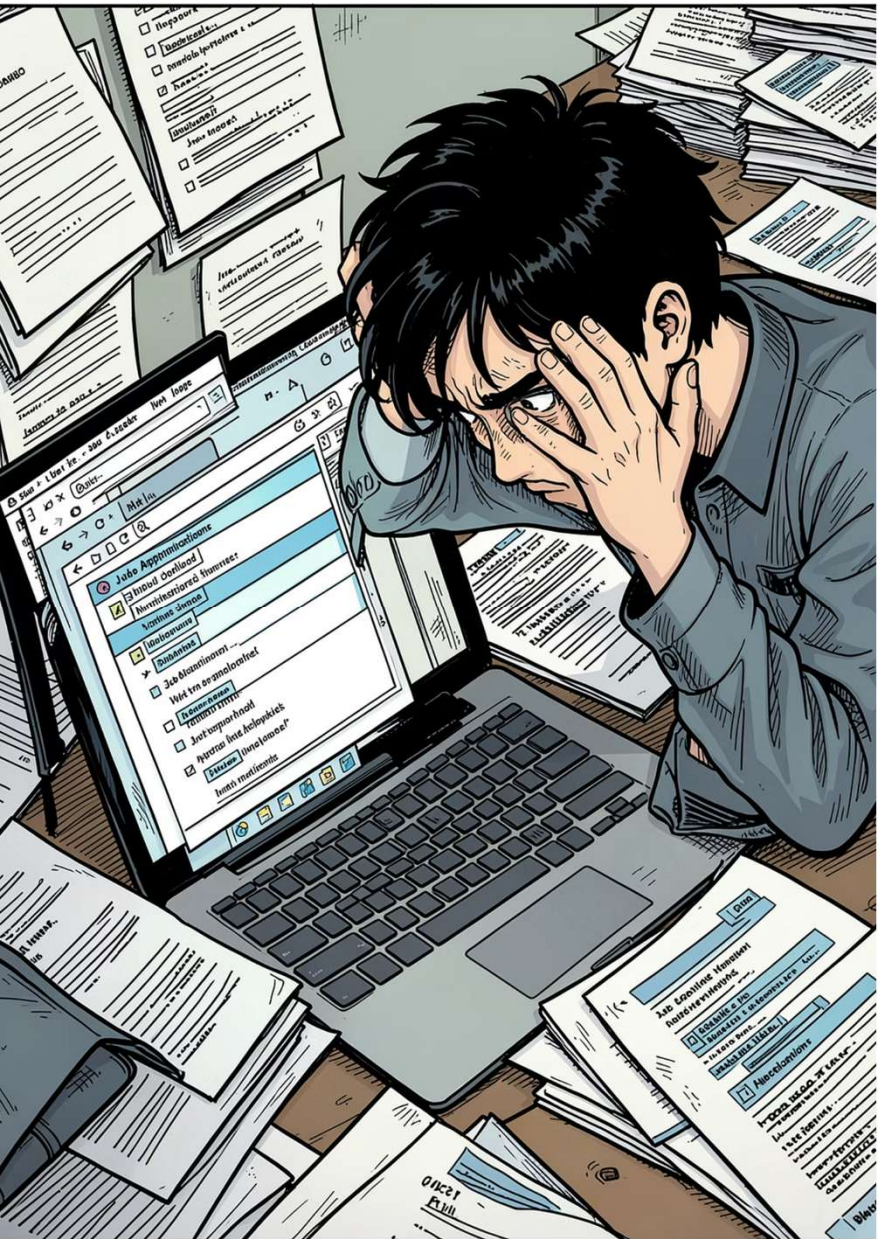


PathMate.ai

AI-Powered Career Development Assistant

Final Presentation · Dohyeon Lee & Heesang Kim · SUNY Korea

https://career-helper-dun.vercel.app



THE PROBLEM

The Career Preparation Process is Fragmented

Students juggle disconnected tools for resumes, cover letters, and interview prep — with no unified guidance.

Resume Creation

Manual effort with no tailored feedback.

Cover Letters

Every application requires a fully custom letter from scratch.

Interview Practice

Real practice opportunities are scarce and hard to access.

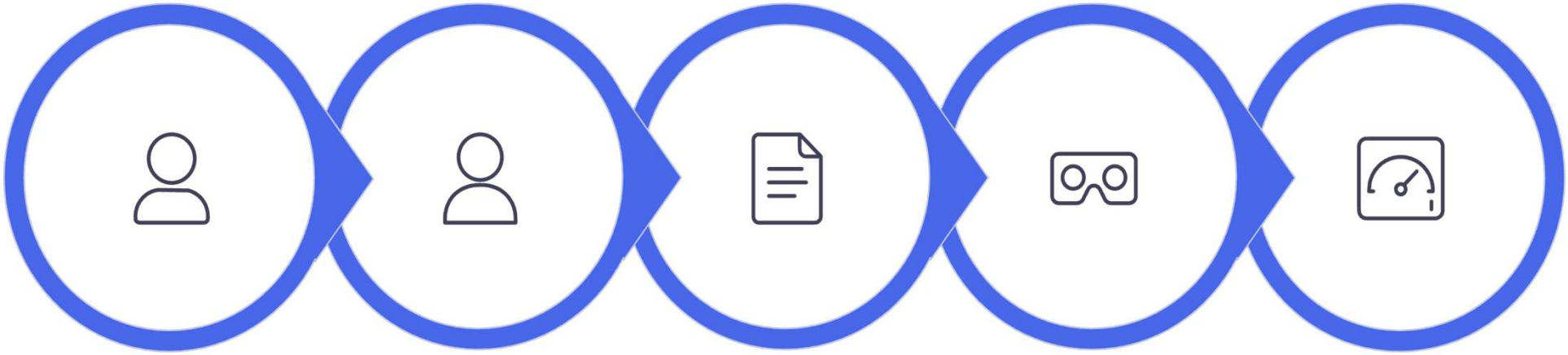
Job Fit Analysis

No clear way to assess alignment with target roles.

OUR SOLUTION

One Integrated Career Development Platform

PathMate.ai consolidates every stage of career preparation into a single AI-powered platform — from profile building to interview readiness.



Create Profile

AI Job Fit

Generate Cover

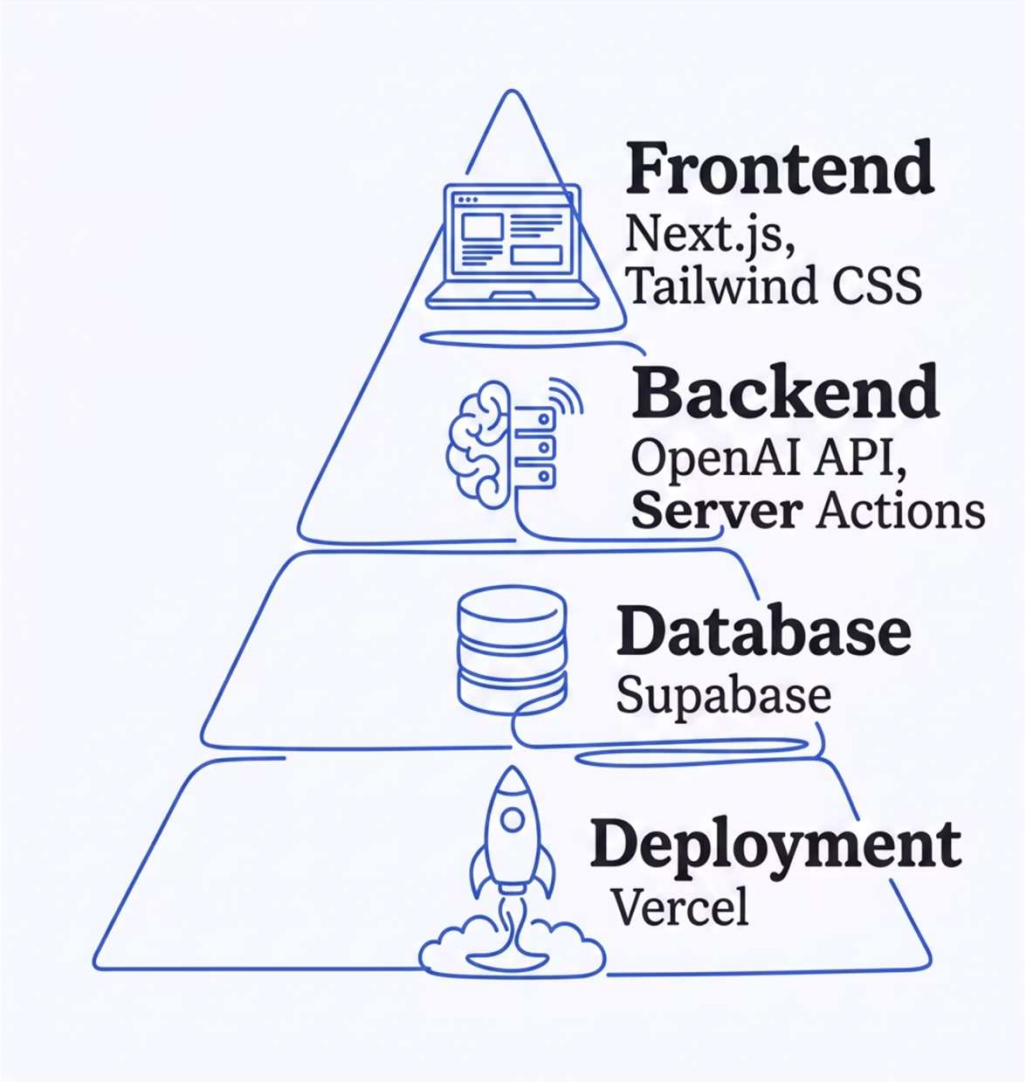
Hiring Simulator

Career Dashboard

From onboarding to career growth, every step is powered by intelligent, personalized AI guidance.

ARCHITECTURE

System Architecture



Tech Stack

- **Frontend:** Next.js + Tailwind CSS
- **Backend:** OpenAI API + Server Actions
- **Database:** Supabase
- **Deployment:** Vercel

A clean, serverless architecture ensures fast performance and seamless AI integration at every layer.

DATA DESIGN

Database Design

Three core entities power PathMate.ai's data layer, linking user profiles to AI analysis and career tracking.

User Profile

- Personal information
- Skills & experience
- Career interests

Interview Sessions

- AI-generated questions
- User responses
- Personalized feedback

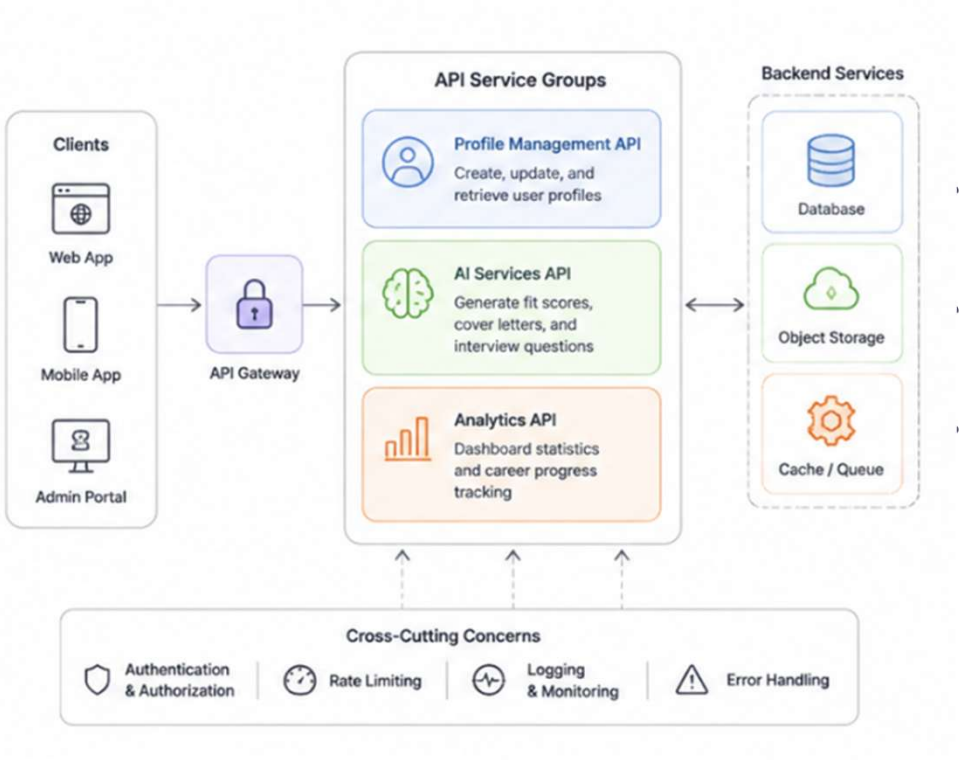
Career Analytics

- AI Fit Scores
- Progress tracking
- Performance trends

BACKEND

Backend API Design

Three API service groups handle all platform functionality — from profile management to AI generation and analytics.



API Service Groups

- **Profile Management:** Create, update, and retrieve user profiles
- **AI Services:** Generate fit scores, cover letters, and interview questions
- **Analytics:** Dashboard statistics and career progress tracking

Each service group operates through secure, well-defined endpoints for a reliable and scalable backend.

FUTURE WORK

Planned Improvements

Here's how PathMate.ai will evolve to better support your career journey:



Resume Parsing

Automatic PDF Resume Analysis



Advanced Personalization

Long-term Career Tracking

Personalized Growth Recommendations



Enhanced AI Features

RAG-based Career Guidance

Industry-Specific Interview Coaching



External Integration

Real Job Posting APIs

LinkedIn Integration

Our vision is to become a comprehensive AI-powered career companion throughout a user's professional journey.



TEAM

Team Contributions

Here's how our team brought PathMate.ai to life:

Dohyeon Lee

- Frontend Development
- UI Component Library
- Dashboard Development
- Profile Management
- Hiring Simulator Interface
- Poster Design

Heesang Kim

- Backend Development
- API Design
- Supabase Integration
- OpenAI Integration
- Deployment
- System Optimization

Shared Responsibilities

- System Integration
- Testing
- Documentation
- Final Deployment

Let’s see our website

PathMate.ai — Your AI-Powered Career Companion